

EthernaCare

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FACULTY OF COMPUTING AND
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Faculty of Computing and Information Technology
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Declaration

The project submitted herewith is a result of my own efforts in totality and in every aspect of the project works. All information that has been obtained from other sources had been fully acknowledged. I understand that any plagiarism, cheating or collusion or any sorts constitutes a breach of TAR University rules and regulations and would be subjected to disciplinary actions.

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Abstract

EthernaCare is a safety-support mobile application developed to address delayed emergency or death discovery among older adults and other individuals who live alone. This issue is becoming more important as Malaysia moves towards an ageing society and more individuals live separately from their family members. When a person living alone experiences an emergency and is unable to request assistance, the situation may remain unnoticed for a long period. EthernaCare aims to reduce this delay while also helping users prepare and securely communicate their post-death preferences.

The application uses a gamified virtual companion named Oren to encourage regular interaction. Instead of using a fixed daily check-in, each user configures a rolling inactivity threshold from one to 168 hours. The first missed threshold window makes the check-in overdue and creates a local reminder. At the second missed window, the server worker attempts an SMS to the user's verified phone. At the third missed window, it records one inactivity alert and attempts one escalation SMS to the verified primary trusted contact. The cloud worker runs every 15 minutes and stores idempotent delivery state, while Android WorkManager provides best-effort local reminders. Oren also includes energy decay, feeding, playing, virtual tokens, owned toys, a shop and weather-based backgrounds. Check-in streaks separately allow users to collect virtual badges and vouchers.

EthernaCare also provides trusted-contact management, emergency location sharing, AI-assisted general guidance and protected Legacy Planning. The Legacy feature allows users to record funeral preferences, Legacy Notes and private documents. After 90 days without a successful check-in, the owner receives a warning and has 24 hours to check in or cancel the process. Otherwise, the verified primary contact may receive protected access for seven days.

The system was developed using Flutter and Supabase with Row Level Security, private Storage, Realtime, Edge Functions and scheduled server processing. Biometric session locking protects a locally restored signed-in session on supported devices. A skippable eight-step first-run guide and a replayable Settings area improve onboarding and accessibility. External SMS, email, weather and AI providers are accessed through controlled service boundaries, and 138 automated tests currently pass with 30.64 percent line coverage.

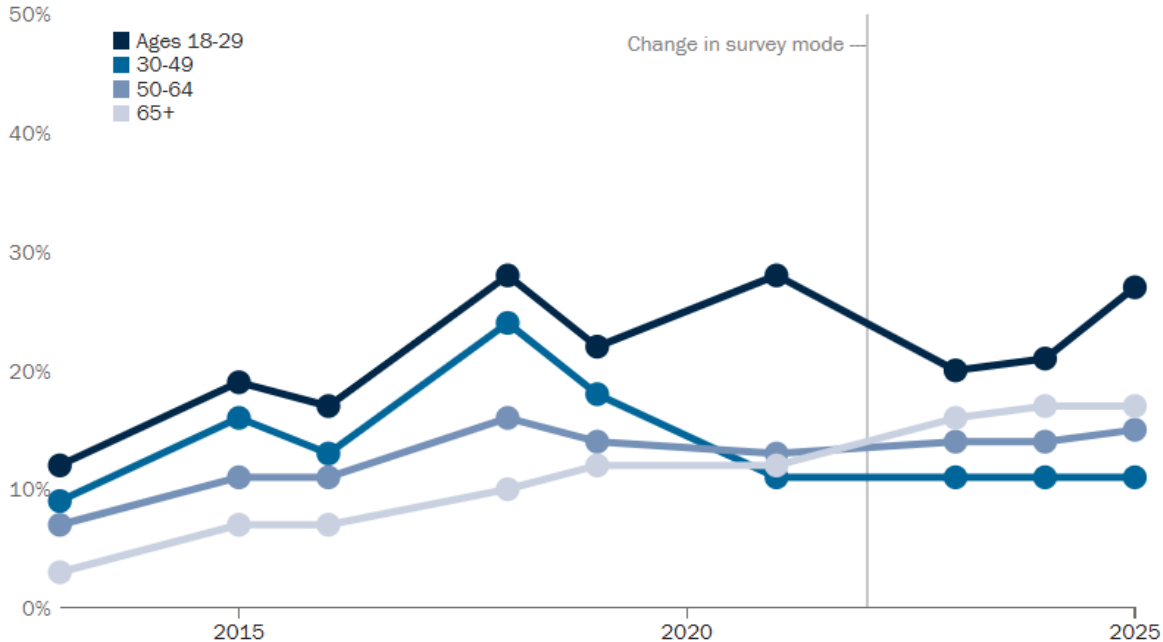
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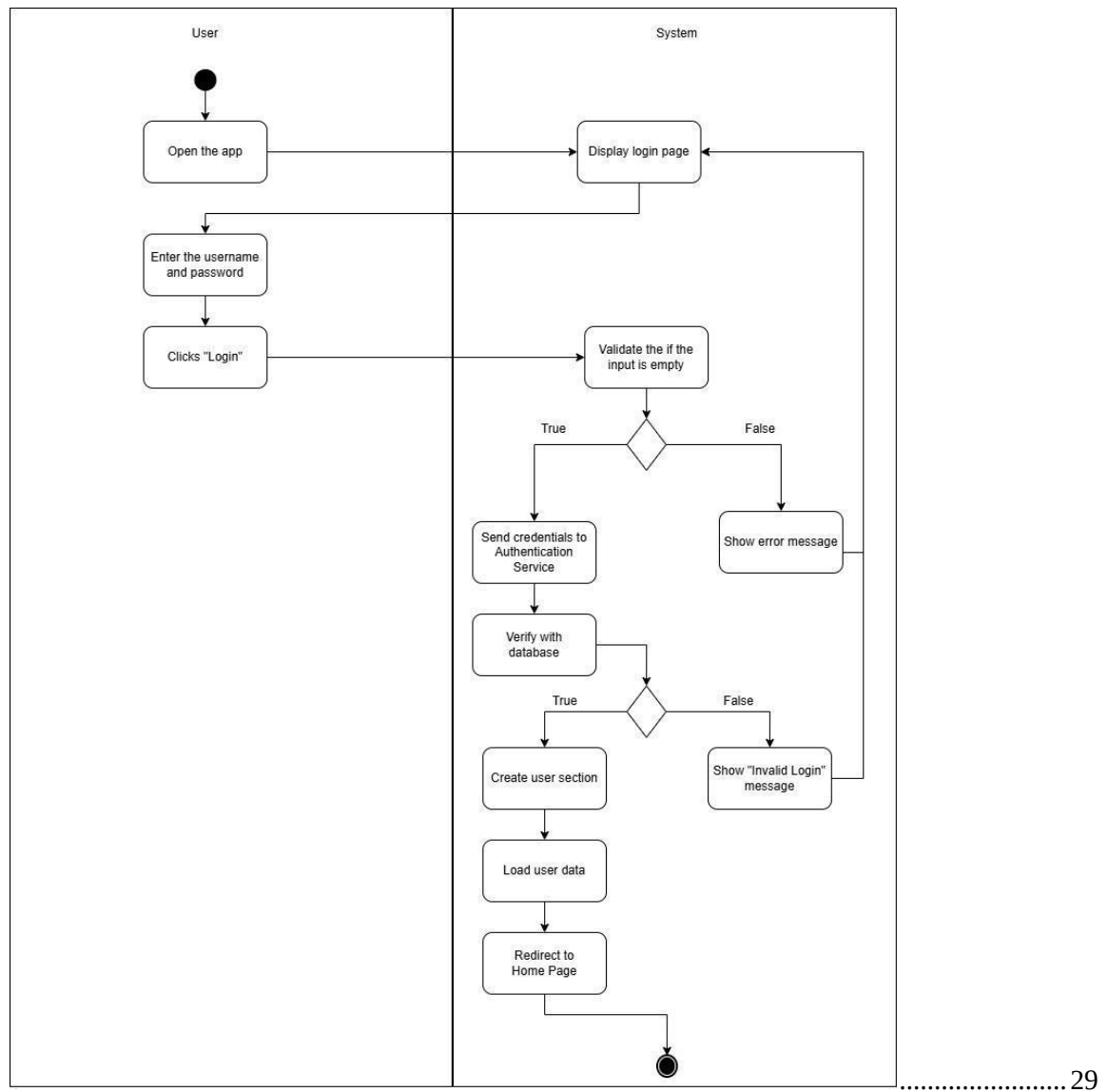
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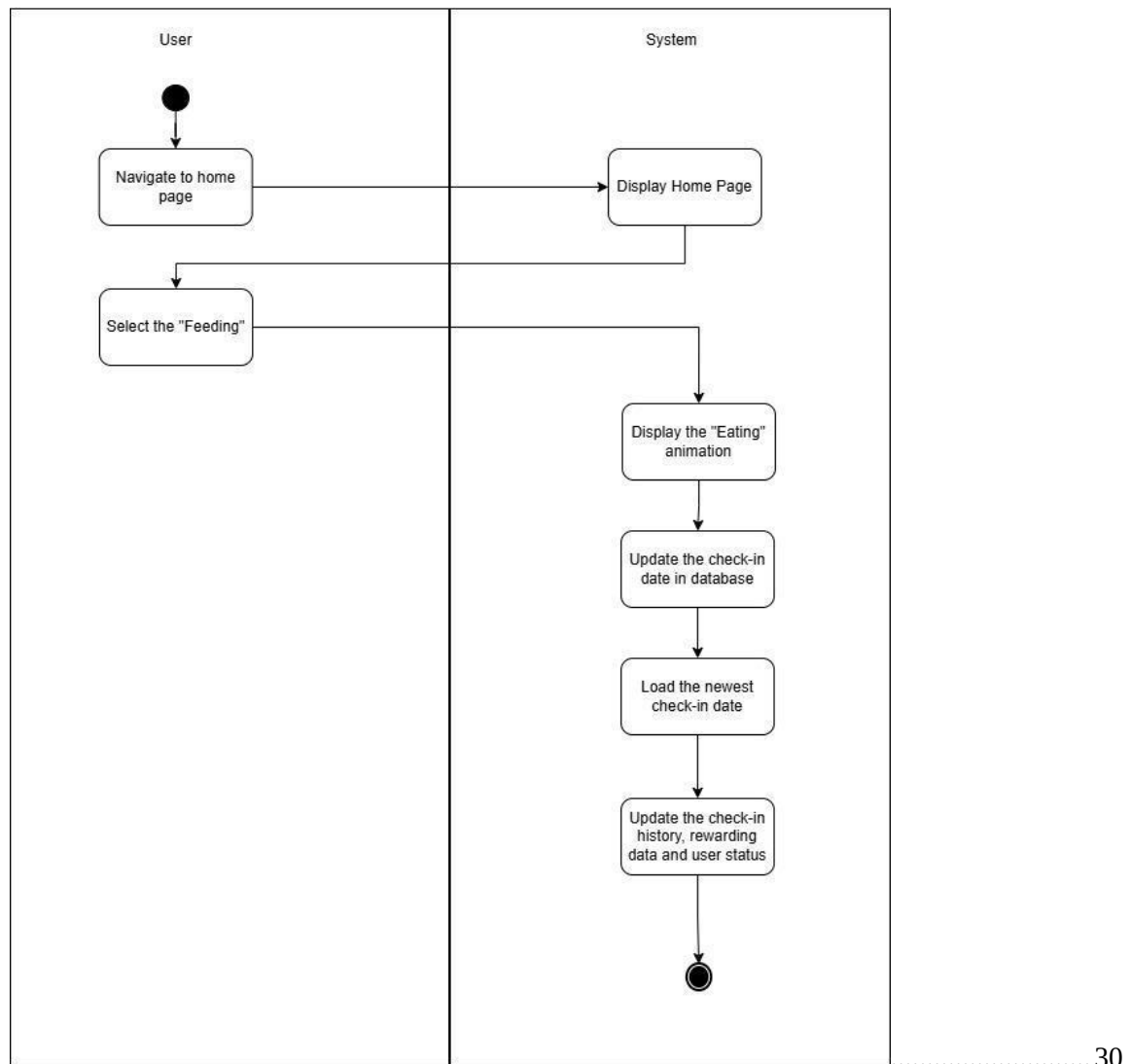
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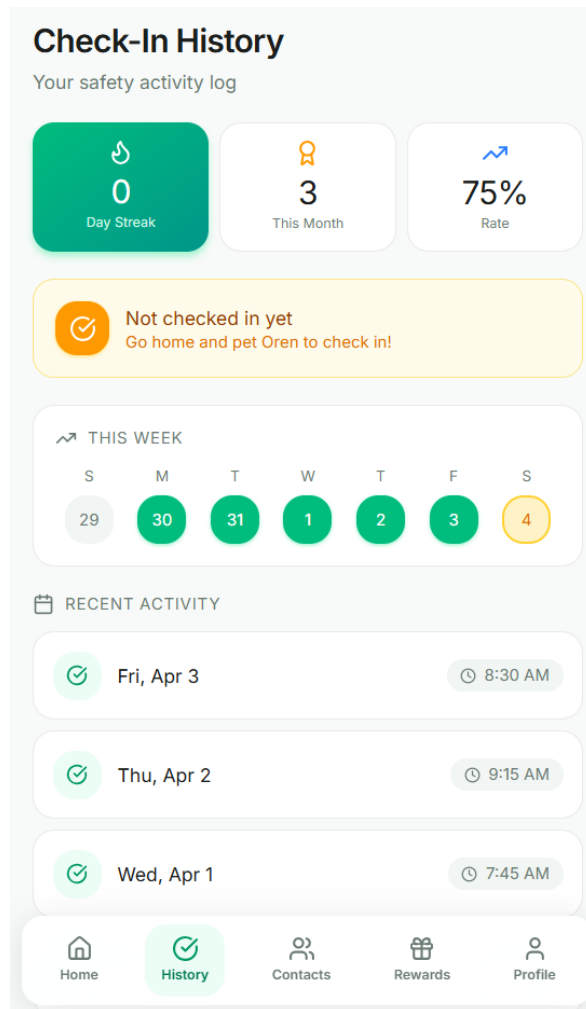
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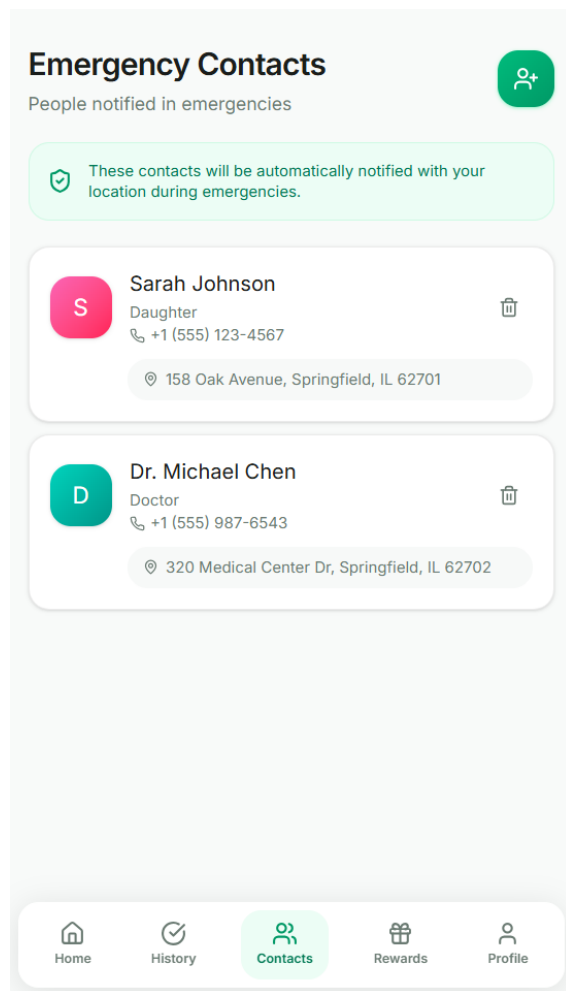
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Chapter 1

Introduction